CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in generative AI systems, from multi-agent LLM pipelines and RAG to containerized deployment, CI/CD and infrastructure as code on Azure and AWS. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in solution architecture and Python development.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning a GenAI knowledge-extraction pipeline end-to-end, designing the solution architecture from data ingestion and chunking to prompt engineering and indexing (agentic, deterministic, and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Containerized the pipeline services with Docker and deployed the Expert Interface on Azure, using LiteLLM for access to LLM APIs.
- Designed and optimized a multi-agent system with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for regulatory and technical documents; validated it against an SAP-system-derived ground truth and raised precision/recall from 33%/45% to 93%/90%.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Exposed the knowledge graph through a Model Context Protocol (MCP) query service with millisecond-latency retrieval, integrated into Claude Code, and built Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve changes.
- **Tech stack: Python, Docker, Kubernetes, Azure, LiteLLM, LangChain, LangGraph, MCP, RAG, GraphRAG, Vector Search, FastAPI, React, Postgres, MongoDB, Neo4j**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, advised on where agentic AI could support them, and translated the requirements into a solution approach.
- Built and deployed an agentic AI application with LangGraph that routes requests across vector database queries, general inference, and summarization, with a React frontend and a Python FastAPI backend tested with Postman.
- Containerized the stack with Docker and deployed it to AWS with infrastructure as code (AWS CDK), with version control and CI/CD in GitLab.
- **Tech stack: Python, Docker, AWS CDK, GitLab CI/CD, AWS (EC2, Lambda), LangChain, LangGraph, FastAPI, React, Postman**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated testing and validation of answer quality and retrieval for the PrismaChat RAG system with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data, and built retrieval pipelines covering chunking strategies, vector databases, indexing, and fine-tuning models for better retrieval.
- Set up AWS infrastructure (Lambda, SQS, DynamoDB) with CDK for the evaluation workflow.

- **Tech stack: Python, PyTorch, Hugging Face Transformers, RAGAS, PytrecEval, LangChain, Azure AI Search, AWS (CDK, Lambda, SQS, DynamoDB)**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data, engineering lag features and rolling statistical windows from sensor data.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, TensorFlow, Scikit-Learn**

PROJECTS

Project: LLM Gateway | Python | FastAPI | Terraform | GCP | BigQuery

- Provider-agnostic gateway over multiple LLM provider APIs with token budgets, streaming, retries, and fallback between providers, logging latency, token usage, and cost for every request; teams onboard self-service by adding a configuration file for their application.
- Deployed to Google Cloud Run with Terraform, with request logs loaded into BigQuery for usage and cost analysis and a KPI dashboard (React, TypeScript); a CI/CD pipeline runs the test suite and an evaluation set on every change.

TECHNICAL SKILLSET

- **Platform & DevOps:** Docker, Kubernetes, CI/CD (GitLab CI/CD), infrastructure as code (AWS CDK), Git, Python microservices
- **Cloud:** Azure (Azure AI Search), AWS (CDK, Lambda, EC2, SQS, S3, DynamoDB, VPC)
- **Generative AI & MLOps:** LiteLLM, LangChain, LangGraph, RAG, GraphRAG, agents and multi-agent systems, MCP, prompt engineering, LLM evaluation (RAGAS, PytrecEval), Hugging Face Transformers, fine-tuning (QLoRA)
- **Programming:** Python, SQL, FastAPI (REST), React, Streamlit
- **Databases:** Postgres, MongoDB, Neo4j, DynamoDB, vector databases (Qdrant, Pinecone, Azure AI Search)
- **Machine Learning:** Scikit-Learn, XGBoost, PyTorch, TensorFlow, time-series analysis
- **Ways of Working:** Agile, Claude Code (custom skills, agentic coding workflows)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)